Comparative Evaluation of Deep Learning Models

for Road Extraction from CartoSat-3 Satellite Imagery

Technical Report

Models: DeepLabV3+ (ResNet-101+scSE) | DLinkNet34 | UNet (DenseNet169+scSE) | LinkNet (DenseNet169) | FPN (DenseNet169+scSE)

Validation: Delhi - 3,446 tiles (cross-city generalisation benchmark)

1. Introduction

Automated extraction of road networks from high-resolution satellite imagery is one of the most operationally important problems in remote sensing and geospatial intelligence. Reliable, up-to-date road maps are indispensable for urban planning, transportation analysis, disaster response and relief coordination, military reconnaissance, infrastructure monitoring, and rapid cartographic updating in newly developing or affected regions. Manual delineation of roads from imagery at the scale and revisit frequency that modern Earth observation missions provide is operationally infeasible. Deep learning based semantic segmentation has therefore emerged as the primary approach for scalable, automated road mapping directly from satellite data.

This report presents a comprehensive comparative evaluation of five distinct deep learning segmentation architectures applied to the problem of binary road extraction from CartoSat-3 imagery. CartoSat-3 is ISRO's highest-resolution earth observation satellite providing 0.28 m panchromatic (PAN) and 1.10 m multispectral (MX) imagery. Both PAN and MX data were utilised in this project - the MX bands provide spectral information across Blue, Green, Red and Near-Infrared channels enabling spectral index computation, while the PAN band at 0.28 m resolution provides the finest spatial detail. The dataset was built using imagery from four Indian cities, all at 1.10 m multispectral resolution, making it one of the highest resolution road segmentation datasets used in published work on Indian urban scenes.

The five models evaluated span a broad design space: DeepLabV3+ with ResNet-101 uses atrous convolution with spatial pyramid pooling for multi-scale context aggregation; DLinkNet34 uses a dilated central block specifically motivated by the linear geometry of roads; UNet with DenseNet169 uses symmetric encoder-decoder architecture with concatenation-based skip connections; LinkNet with DenseNet169 uses a lightweight residual decoder with additive skip connections; and FPN with DenseNet169 uses a top-down feature pyramid for multi-scale lateral fusion. Three of the five share the DenseNet169 encoder backbone, enabling controlled comparison of decoder design choices. All models except DLinkNet34 also incorporate scSE attention mechanisms in the decoder.

2. Problem Formulation

The road extraction task is formulated as a binary semantic segmentation problem. For every input satellite image tile of spatial dimensions H x W with C spectral channels, the model must predict a pixel-wise binary road mask of the same spatial dimensions. Each pixel is independently classified as either road (positive class, label 1) or non-road (negative class, label 0).

Formal Problem Statement:

Given: X in R^{H x W x C} (input satellite tile, C = 5 for RGB+NDVI+NDWI)
Estimate: P(i,j) = P(road at pixel (i,j) | X) in [0, 1] via sigmoid activation

Binary prediction via threshold tau = 0.50:

```
Y(i,j)  =  1   if  P(i,j) > 0.50        (road pixel)
Y(i,j)  =  0   otherwise                  (non-road pixel)
```

Road extraction is technically challenging for several interconnected reasons. The road class is geometrically diverse - roads range from narrow residential lanes of 3-4 metres to broad national highways of 24 metres or more. Roads are linear and highly connected structures, meaning a locally correct but globally disconnected prediction (fragmented road segments) is practically useless for navigation or mapping. The spectral appearance of road surfaces overlaps significantly with other impervious surfaces including rooftops, parking areas, concrete plazas, and barren ground, creating persistent sources of false positives. Shadows from tall buildings in dense urban areas can completely occlude road segments. Vegetation overhanging roads is a major challenge in semi-urban and tree-lined areas. The class imbalance between road pixels and background is severe - road pixels typically constitute only 5-15% of pixels in urban tiles.

3. Dataset Description and Preparation

The dataset was constructed entirely from CartoSat-3 multispectral satellite imagery and corresponding road labels derived from OpenStreetMap annotations processed through a comprehensive geospatial preprocessing pipeline to ensure pixel-level accuracy between image tiles and ground-truth masks.

3.1 Dataset Statistics and Split Design

3.1.1 Cross-City Dataset Breakdown

Split	Cities	Tile Count	Purpose
Training	Ahmedabad + Hyderabad + Ludhiana	10,509 pairs	Model learning
Validation	Delhi (unseen city)	3,446 pairs	Generalisation benchmark
Total	4 cities	13,955 pairs	Complete dataset

The deliberate city-disjoint split means the validation set represents a genuine out-of-distribution test. Delhi has significantly different urban morphology from the training cities: its road network includes wide planned arterials in New Delhi alongside extremely dense unplanned layouts in older areas, a mix rarely seen in Ahmedabad, Hyderabad, or Ludhiana. The model achieves the reported scores without ever having seen a single pixel of Delhi during training.

The training set comprises 10,509 image-mask tile pairs drawn from three geographically and morphologically distinct Indian cities. The validation set consists of 3,446 pairs from Delhi, a city never seen during training. All tiles are 512 x 512 pixels. Images are stored as UInt16 GeoTIFF with 4 bands (B, G, R, NIR). Masks are stored as Byte (0/1) GeoTIFF. The cross-city split is the most critical design decision in this dataset: validation on an entirely unseen city provides a genuine test of generalisation rather than in-domain overfitting.

Training/Validation ratio: 10,509 / 3,446 = 75% / 25% by tile count. The three training cities were chosen to cover diverse Indian urban morphologies: Ahmedabad represents mixed urban-semi-urban zones with transitional road widths; Hyderabad represents dense historical-core layouts with narrow and irregular lanes; Ludhiana represents the planned grid-pattern cities common in north India. Delhi in validation represents the most complex morphological mix, including modern planned areas (Lutyen's Delhi, Dwarka) alongside extremely dense unplanned settlements (Shahdara, Chandni Chowk), making it a rigorous generalisation test.

3.1.2 Standardized Dataset Breakdown

The dataset was split at the image-mask pair level using a city-stratified random split strategy. Rather than shuffling all tiles from all cities together (which could cause city-level imbalance across splits), the split was performed independently within each city first, then the per-city subsets were merged into the final train and validation folders. This ensures that every city contributes proportionally to both splits, and neither the training set nor the validation set is dominated by any single city.

Split Method

The procedure followed these steps, applied identically to each of the four cities:

- Image-mask pairs are identified from filenames. For example, ahmedabad_01_01.tif is always paired with ahmedabad_mask_01_01.tif. The tile ID (row_col) is the pairing key -- image and mask are never separated.
- Within each city, all paired tiles are shuffled using a fixed random seed for reproducibility.
- Approximately 70% of each city's tiles are assigned to training; 30% are assigned to validation.
- The four city-wise training subsets are merged into one final train folder. The four city-wise validation subsets are merged into one final val folder.
- The original test partition was merged with the validation partition to create a single larger 30% validation split, giving a more stable validation metric during model development.

The resulting split guarantees that samples from all four cities are present in both training and validation. This is a form of stratification by city, preventing a scenario where one city appears only in training or only in validation. The table below shows the per-city contribution to each split:

City	Total tiles	Training (70%)	Train images	Validation (30%)	Val images
Ahmedabad	3532	70% of city tiles	2472	30% of city tiles	1060
Hyderabad	3607	70% of city tiles	2525	30% of city tiles	1,082
Ludhiana	3370	70% of city tiles	2359	30% of city tiles	1011
Delhi	3446	70% of city tiles	2412	30% of city tiles	1034
TOTAL (merged)	13955	All cities combined	9,768 images	All cities combined	4,187 images

Final Dataset Counts

After merging the per-city splits, the confirmed image and mask counts are:

Split	Images	Masks	Composition
Training	9,768	9,768	70% of each city - Ahmedabad, Hyderabad, Ludhiana, Delhi
Validation	4,187	4,187	30% of each city - Ahmedabad, Hyderabad, Ludhiana, Delhi
Total	13,955	13,955	All four cities represented in both splits

3.2 Imagery Specification - PAN and MX Data

CartoSat-3 delivers two complementary image products for each acquisition. The panchromatic (PAN) band captures a broad visible wavelength range at 0.45 m ground sampling distance. The multispectral (MX) product delivers four spectral bands using imagery at 0.45 m PAN and 1.10 m MX resolution. Both products were used in building this dataset.

Property	PAN Band	MX Bands (B, G, R, NIR)
Spatial Resolution	0.45 m GSD	1.10 m GSD
Spectral Range	450-900 nm (broadband)	450-860 nm (4 discrete bands)
Bit Depth	UInt16 (16-bit)	UInt16 (16-bit)
Primary Use	Spatial detail, pan-sharpening reference	Spectral indices (NDVI, NDWI), RGB composite
Tile size after retiling	512 x 512 pixels	512 x 512 pixels
Mask data type	N/A (reference only)	Byte (0/1 binary road label)

The MX bands at 1.10 m were used as the primary input for the segmentation models since they provide the spectral richness needed for NDVI and NDWI computation. The PAN band at 0.45 m is substantially finer than most public road segmentation datasets (which typically use 1-2 m resolution imagery), enabling detection of narrow residential lanes and lane markings that would be sub-pixel at coarser resolutions.

3.3 Ground-Truth Mask Generation

Ground truth road masks were generated from OpenStreetMap road centreline data through a carefully designed geospatial pipeline addressing three specific challenges. First, a GPS positional offset between OSM vectors and actual road pixels in the satellite image was corrected by translating the road vectors by +3 m East and +12 m North in UTM metric space using the aligning.py script. Second, zero-width centrelines were converted into filled road-width polygons through hierarchical buffering

based on highway classification using the buffering.py script (trunk 12 m, primary 8 m, secondary 6 m, tertiary 4 m, residential 3 m, unclassified 2 m half-widths). Third, careful rasterisation matched the exact pixel grid of the CartoSat-3 imagery. After tiling, tiles with empty or near-empty masks were removed to eliminate non-informative training samples.

3.4 Complete Dataset Creation Pipeline

The dataset was built from scratch through an 11-step geospatial processing pipeline. Each step addresses a specific challenge in converting raw satellite data and crowd-sourced road annotations into a clean, aligned, pixel-accurate training corpus. The complete flow is illustrated below, followed by detailed explanation of each step with formulas and code.

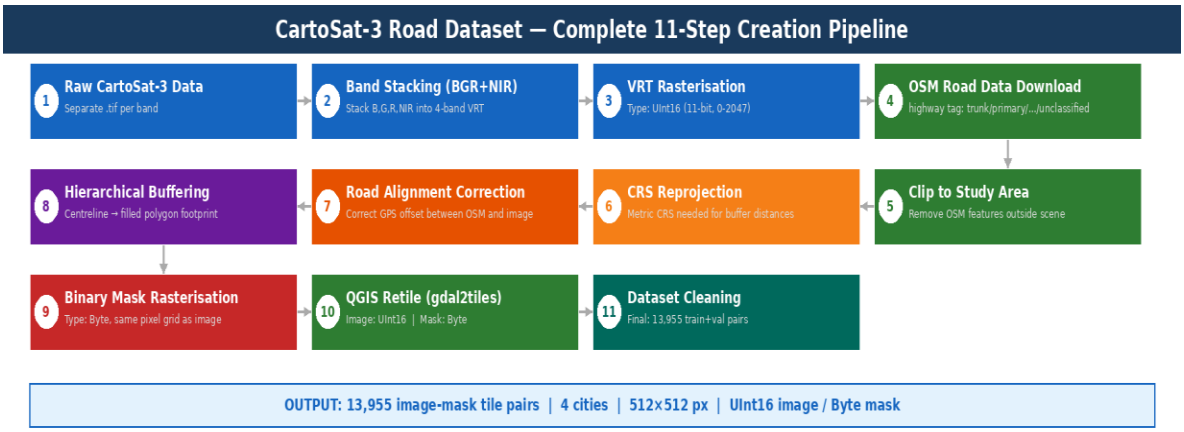


Figure: Complete 11-step dataset creation pipeline from raw CartoSat-3 bands to cleaned 512x512 tile pairs.

Step 1-3: Band Stacking and VRT Rasterisation

CartoSat-3 delivers four spectral bands (Blue, Green, Red, NIR) as separate UInt16 GeoTIFF files. These are first combined into a single 4-band Virtual Raster Table (VRT) using gdal_buildvrt, then exported to a final 4-band GeoTIFF using gdal_translate. The band order in the output stack is Band 0 = Blue, Band 1 = Green, Band 2 = Red, Band 3 = NIR. This order is consumed directly by compute_input_features() during training. The UInt16 data type (11-bit data, 16-bit storage, range 0-2047) is preserved throughout -- downscaling to 8-bit at this stage would discard 99.6% of the available radiometric dynamic range.

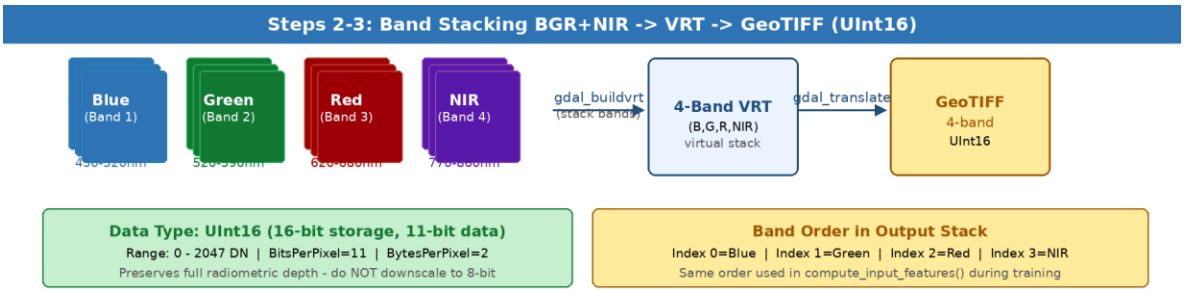


Figure: Band stacking workflow -- four individual band files combined via VRT into a single 4-band UInt16 GeoTIFF.

Step 4-5: OSM Road Data and Area Clipping

Road reference data is obtained from OpenStreetMap (OSM) via the Overpass API. OSM provides vector road centrelines with the 'highway' attribute tag encoding road classification (trunk, primary, secondary, tertiary, residential, unclassified, service). The downloaded shapefile is clipped to the spatial extent of the CartoSat-3 imagery to ensure that road annotations only exist where satellite coverage is available, preventing spurious mask values at tile borders.

Step 6: CRS Reprojection to UTM

All metric-distance operations (alignment correction and buffering) must be performed in a projected coordinate system where units are metres, not degrees. The OSM road data is reprojected from geographic coordinates (EPSG:4326, WGS84 decimal degrees) to UTM Zone 43N (EPSG:32644 / EPSG:32643) before any geometric processing. This ensures that buffer distances specified in metres produce geometrically accurate results rather than degree-based approximations that vary with latitude.

Step 7: Road Alignment Correction -- aligning.py

A systematic spatial offset was identified between the OSM road centrelines and the actual road pixels visible in the CartoSat-3 imagery. This offset arises from GPS positional errors in crowd-sourced OSM data combined with the satellite's 3.2-degree tilt angle. The offset is corrected by applying a rigid translation to all road geometries in UTM metric space. The translation vector was determined empirically by visual overlay inspection across multiple tiles.

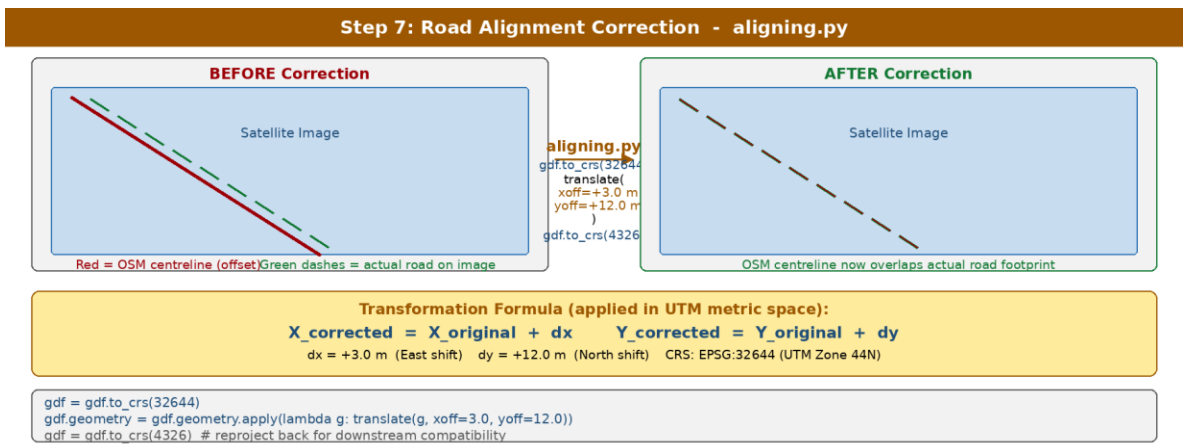


Figure: Road alignment correction applied by aligning.py -- OSM centreline shifted +3m East, +12m North in UTM space.

Step 8: Hierarchical Road Buffering -- buffering.py

OSM road data provides zero-width centrelines. For semantic segmentation, the mask must represent the full road width as a filled polygon. A variable-width buffer is applied to each centreline based on the OSM highway classification, reflecting real-world road widths. The buffering is applied in UTM metric space (after automatic reprojection if the input CRS is geographic), ensuring buffer distances in metres are geometrically accurate. The key function from buffering.py:

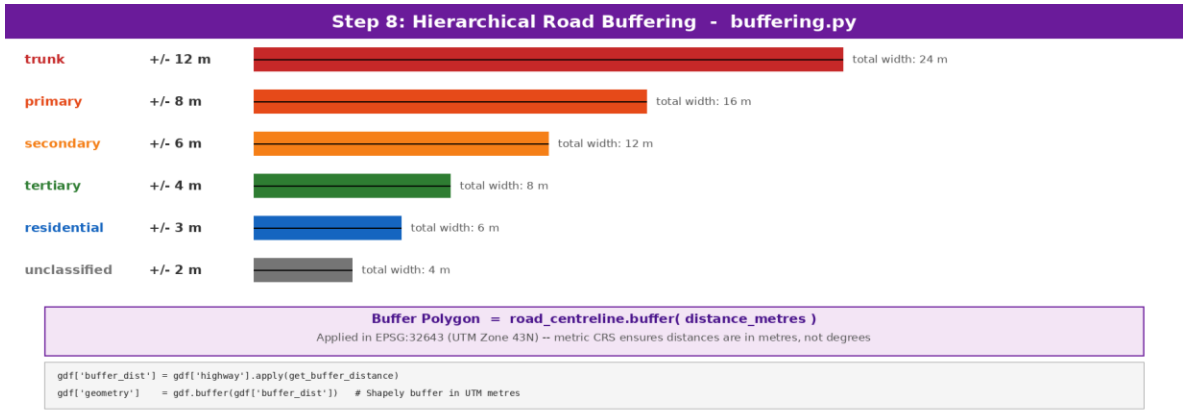


Figure: Hierarchical buffer distances by road type from buffering.py -- each class receives a width proportional to its real-world size.

Step 9: Binary Mask Rasterisation

The buffered road polygon Shapefile is rasterised onto the same pixel grid as the CartoSat-3 imagery using GDAL rasterize. Pixels whose centres fall within a buffered road polygon are assigned value 1 (road); all others receive value 0 (background). Output data type is Byte (uint8), values 0 or 1 only. The rasterisation matches exactly the spatial extent, resolution, and CRS of the image raster, ensuring pixel-perfect alignment between image tiles and mask tiles.

Steps 10-11: QGIS Retile and Dataset Cleaning

Full-scene GeoTIFFs (covering entire city areas) are subdivided into 512x512 pixel tiles using the QGIS Retile function (internally gdal2tiles). Image tiles are stored as UInt16 4-band GeoTIFF. Mask tiles are stored as Byte (uint8) single-band GeoTIFF with values 0 or 1. Tiles are named city_row_col.tif (images) and city_mask_row_col.tif (masks) for deterministic pair matching. After tiling, a cleaning pass removes: (a) tiles where the mask is entirely zero (no road pixels -- no learning signal); (b) near-empty mask tiles below a minimum road fraction; (c) border tiles with large NoData regions. This yields the final 10,509 training and 3,446 validation tile pairs.

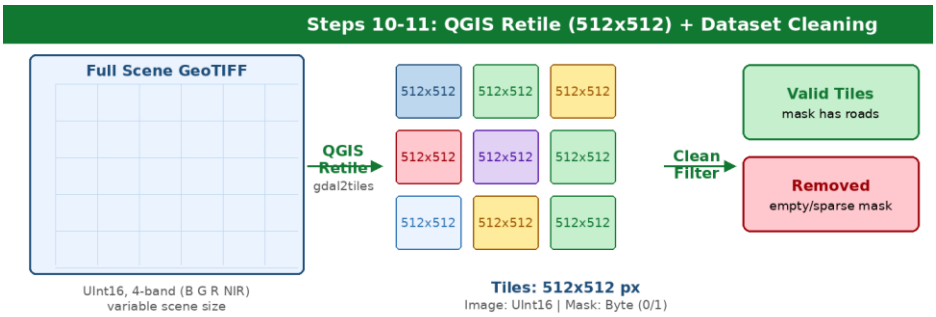


Figure: QGIS Retile into 512x512 tiles and cleaning pass to remove empty/sparse mask tiles. Final split: 10,509 train + 3,446 val.

4. Input Representation and Spectral Feature Engineering

A key design decision in this pipeline is the use of derived spectral indices in addition to raw RGB channels. Rather than providing only the three visible bands to the model, two physically meaningful spectral indices are computed from the NIR band and appended to form a five-channel input tensor.

4.1 NDVI - Normalised Difference Vegetation Index

$$\text{NDVI} = \frac{\text{NIR} - \text{R}}{\text{NIR} + \text{R} + \text{epsilon}}$$

epsilon = 1x10^-6 Range: -1.0 to +1.0 High: dense vegetation Low: roads, bare soil

Roads, being impervious surfaces with no photosynthetic activity, exhibit very low NDVI values (typically -0.1 to +0.2). Vegetation exhibits high NDVI (+0.3 to +0.8). Including NDVI as an input channel gives the model an explicit discriminative signal that distinguishes road pixels from vegetation pixels even where RGB texture alone would be ambiguous. This is especially important in semi-urban scenes where road surfaces partially overlap in RGB colour space with soil or grass, and in tree-lined avenues where canopy edges partially overlap road edges.

4.2 NDWI - Normalised Difference Water Index

$$\text{NDWI} = \frac{\text{G} - \text{NIR}}{\text{G} + \text{NIR} + \text{epsilon}}$$

epsilon = 1x10^-6 Range: -1.0 to +1.0 High: open water Low/negative: roads, buildings

Water bodies such as rivers, canals, ponds, and flooded areas can appear visually similar to roads in RGB imagery, particularly under low-sun conditions or when imaged at nadir. NDWI provides strong contrast between water (high positive NDWI) and roads (low or negative NDWI), suppressing false positive detections adjacent to water bodies. All four cities in this dataset - Delhi, Hyderabad, Ahmedabad, and Ludhiana - contain significant water features that would otherwise cause persistent confusion.

4.3 Final 5-Channel Input Tensor

Input = [B, G, R, NDVI, NDWI] Shape: (5, 512, 512) dtype: float32

Fallback (if NIR unavailable): [B, G, R] -> (3, 512, 512)

4.4 Dataset-Specific Normalisation

Normalisation statistics are computed from the training set itself rather than borrowed from ImageNet statistics. ImageNet statistics are derived from natural 8-bit RGB photographs and are fundamentally inappropriate for 16-bit multispectral satellite data with physically different spectral channels and intensity ranges.

Per-channel statistics computed over all training pixels (up to 200 sampled images):

```
    mu_c   =  (1/N) * sum_i  x_{i,c}
    sigma_c  =  sqrt( (1/N) * sum_i  x_{i,c}^2  -  mu_c^2 )
    if sigma_c < 1e-6  then  sigma_c = 1.0    <- stability guard
prevents division by near-zero
```

Applied at runtime to every tile: **X_c_norm = (X_c - mu_c) / sigma_c**

Identical normalisation applied during training, validation, and single-sample inference.

5. Shared Training Pipeline

All five models were trained using an identical training pipeline, ensuring that observed performance differences are attributable to architectural differences alone. This controlled experimental design is essential for a valid architectural comparison.

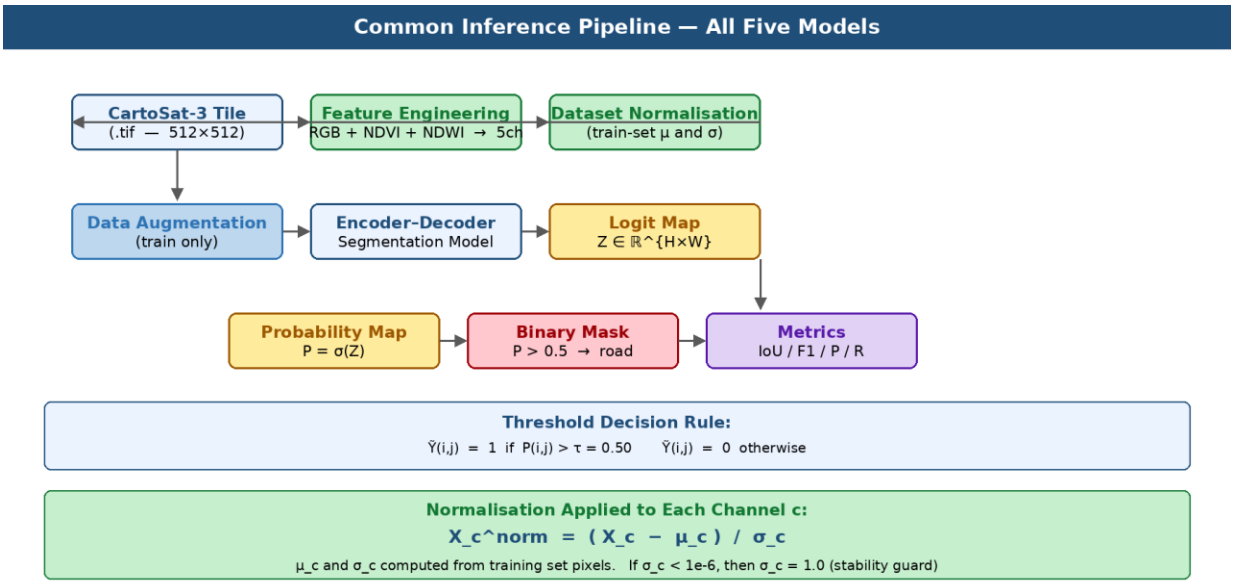


Figure 1: Common inference pipeline applied to all five models. Training additionally includes the data augmentation step shown in Section 5.1.

5.1 Data Augmentation - Full Parameters from Code

Augmentation is applied only during training through the Albumentations library. Both image and paired mask receive identical geometric transformations to preserve pixel-level alignment. The exact parameters used in the implementation are listed below:

Augmentation	Class	Key Parameters	Probability	Rationale
Horizontal Flip	A.HorizontalFlip	(default)	p=0.50	Road networks are symmetric across horizontal axis
Vertical Flip	A.VerticalFlip	(default)	p=0.50	Adds north/south orientation invariance

Augmentation	Class	Key Parameters	Probability	Rationale
Random Rotate 90	A.RandomRotate90	(default)	p=0.50	Covers all four cardinal orientations (0, 90, 180, 270 deg)
Transpose	A.Transpose	(default)	p=0.20	Axis-swap diversity, simulates diagonal view
Shift Scale Rotate	A.ShiftScaleRotate	shift_limit=0.03, scale_limit=0.05, rotate_limit=10, interpolation=INTER_LINEAR, border_mode=BORDER_REFLECT_101	p=0.40	Sensor angle and tile framing variation
Brightness Contrast	A.RandomBrightnessContrast	brightness_limit=0.15, contrast_limit=0.15	p=0.30	Atmospheric and illumination condition variation
Gaussian Noise	A.GaussNoise	std_range=(0.01, 0.05), mean_range=(0.0, 0.0)	p=0.20	Sensor noise robustness

Photometric augmentations (RandomBrightnessContrast, GaussNoise) are applied only to the image tensor. Geometric augmentations (HorizontalFlip, VerticalFlip, RandomRotate90, Transpose, ShiftScaleRotate) are applied identically to both image and mask tensors to maintain pixel-perfect correspondence. Augmentation is entirely disabled during validation and inference.

5.2 Loss Function

All models are trained with the same combined Dice plus Binary Cross-Entropy loss function. The Dice component addresses the severe class imbalance between road and non-road pixels by directly optimising region overlap. The BCE component provides stable per-pixel supervision, especially important during early training:

$$L_total = 0.7 \times L_Dice + 0.3 \times L_BCE$$

Dice: $L_Dice = 1 - (2 \times sum_i\ p_i * y_i + eps) / (sum_i\ p_i + sum_i\ y_i + eps)$

eps = 1x10^-6 (smooth parameter) p_i = predicted probability y_i = ground truth

BCE: $L_BCE = -(1/N) * sum_i [y_i*log(sigma(z_i)) + (1-y_i)*log(1-sigma(z_i))]$

z_i = raw logit sigma = sigmoid function N = total pixels in batch

Dice handles class imbalance. BCE stabilises gradient flow in early epochs. 0.7/0.3 weighting prioritises overlap quality.

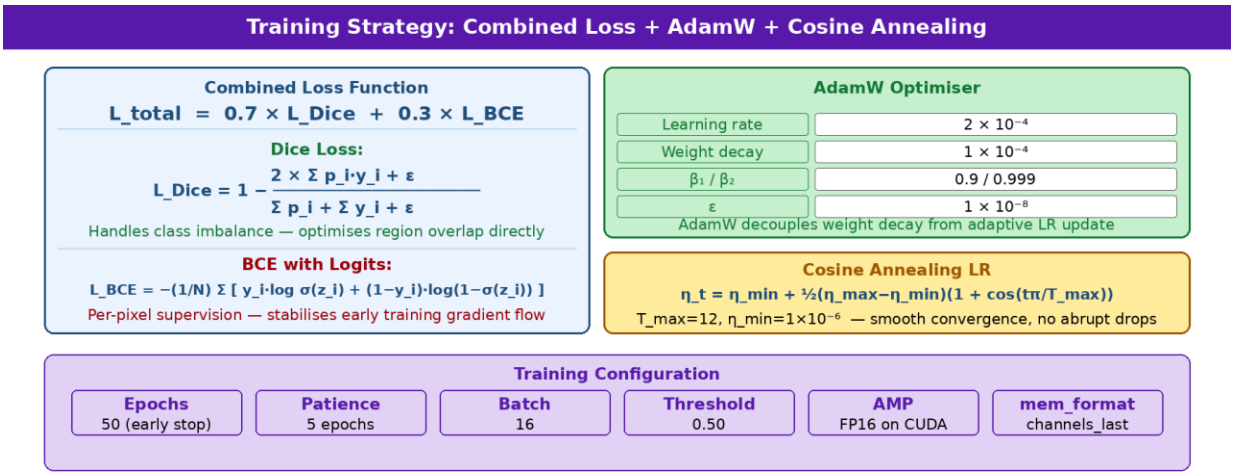


Figure 2: Combined loss function, AdamW optimiser configuration, and cosine annealing LR schedule - identical for all five models.

5.3 Optimiser, Scheduler, and Training Configuration

The following training configuration was used identically for all five models. These parameters were set based on common best practice for satellite image segmentation and kept constant to ensure fair comparison:

Parameter	Value	Code Reference	Notes
Optimizer	AdamW	torch.optim.AdamW	Decoupled weight decay from adaptive LR
Learning Rate	2x10^-4	lr=2e-4	Initial LR for cosine schedule
Weight Decay	1x10^-4	weight_decay=1e-4	L2 regularisation applied to weights
LR Scheduler	Cosine Annealing	CosineAnnealingLR	Smooth decay, no abrupt drops
T_max (scheduler)	12 epochs	T_max=12	Period for one cosine half-cycle
eta_min (scheduler)	1x10^-6	eta_min=1e-6	Minimum LR at cycle trough
Batch Size	16	BATCH_SIZE=16	Per-GPU batch; 16 tiles of 512x512
Max Epochs	50	EPOCHS=50	Upper bound before forced stop

Parameter	Value	Code Reference	Notes
Early Stopping Patience	5 epochs	PATIENCE=5	Stop if Val IoU does not improve
Validation Threshold	0.50	VAL_THRESH=0.50	Sigmoid output threshold for binary mask
Mixed Precision	FP16 + GradScaler	torch.amp.autocast	Reduces GPU memory, speeds training
Memory Format	channels_last	torch.channels_last	Optimises CUDA conv throughput
Random Seed	42	SEED=42	Python, NumPy, PyTorch, CUDA all seeded
Num Workers	2	NUM_WORKERS=2	DataLoader parallel workers
Pin Memory	True	pin_memory=True	Faster CPU-GPU data transfer
Prefetch Factor	2	prefetch_factor=2	Batches preloaded in background

The cosine annealing schedule is mathematically defined as: eta_t = eta_min + 0.5 * (eta_max - eta_min) * (1 + cos(t*pi / T_max)). This provides smooth learning rate decay from 2x10^-4 down to 1x10^-6 over T_max=12 epochs, then resets. The best model checkpoint is saved whenever validation IoU strictly exceeds the previous best, storing the complete training state including model weights, optimizer state dict, scheduler state dict, GradScaler state dict, best IoU value, and best epoch number.

6. Model Architectures - Detailed Description

This section provides a thorough architectural description of each of the five models evaluated. Understanding the structural differences is essential for interpreting performance results and understanding why specific models excel on particular evaluation dimensions. All five models output a single-channel logit map Z of shape (1, 512, 512), converted to a probability map via sigmoid activation P = sigma(Z) = 1 / (1 + exp(-Z)), then thresholded at 0.50 to produce the binary road mask.

6.1 Model 1: DeepLabV3+ - ResNet-101 Encoder + scSE Attention

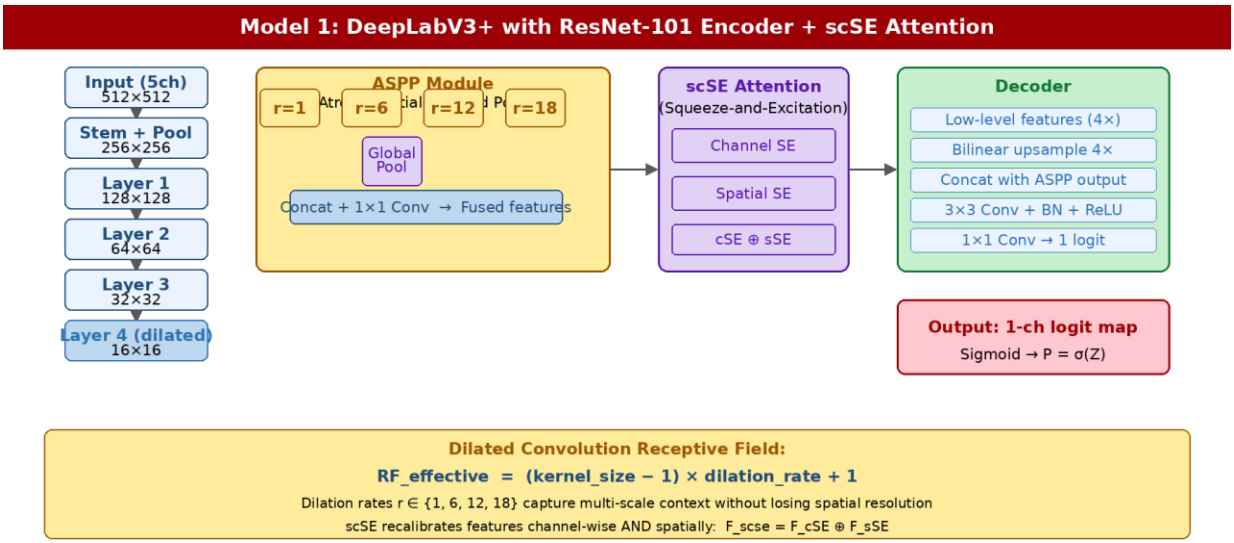


Figure 3: DeepLabV3+ architecture with ResNet-101 encoder, ASPP multi-scale module, and scSE attention in the decoder.

```
Implementation:  smp.DeepLabV3Plus(encoder_name='resnet101',
encoder_weights=None,
                  in_channels=IN_CHANNELS, classes=1, activation=None,
                  decoder_attention_type='scse')
```

Encoder: ResNet-101

ResNet-101 is a 101-layer residual network using residual skip-add connections between blocks. Each residual block computes output = F(x) + x, where F(x) is the block transformation and x is the identity shortcut. ResNet-101's final stages use dilated convolutions to maintain a larger feature map output size (1/16 of input) rather than reducing to 1/32, preserving more spatial information. The encoder produces feature maps at scales 1/4, 1/8, 1/16, and 1/32 of input with channel depths 256, 512, 1024, and 2048 respectively. The first convolution was modified to accept 5 input channels, with extra channels initialised by interpolating from the pretrained 3-channel weights.

ASPP - Atrous Spatial Pyramid Pooling

The ASPP module is the defining component of the DeepLab family. It applies multiple parallel dilated convolutions with different dilation rates to the encoder output, each capturing context at a different scale, then concatenates all branches:

ASPP Branches applied in parallel to encoder output:

```
Branch 1: 1x1 Conv (r=1)  ->  receptive field: 1 pixel
Branch 2: 3x3 Conv, dilation r=6  ->  RF = (3-1)*6+1 = 13 pixels
Branch 3: 3x3 Conv, dilation r=12  ->  RF = (3-1)*12+1 = 25 pixels
Branch 4: 3x3 Conv, dilation r=18  ->  RF = (3-1)*18+1 = 37 pixels
Branch 5: Global Average Pooling  ->  image-level context
All 5 branches concatenated  ->  1x1 Conv to 256 channels
```

Effective RF formula for dilated conv: RF = (kernel_size - 1) x dilation_rate + 1

scSE Attention and Decoder

The scSE (spatial and channel Squeeze-and-Excitation) module applies two complementary attention mechanisms. The channel SE branch computes global average pooling followed by two fully connected layers to produce a channel-wise recalibration vector - asking which feature channels are most informative. The spatial SE branch applies a 1x1 convolution to produce a single-channel spatial attention map - asking which spatial locations are most informative. The outputs combine as: F_scSE = F times cSE(F) + F times sSE(F). The decoder takes the ASPP output, bilinearly upsamples 4x, concatenates with low-level encoder features after a 1x1 Conv, applies 3x3 Conv blocks with scSE, and produces the final 1-channel logit map.

6.2 Model 2: DLinkNet34 - ResNet-34 + Dilated Central D-Block

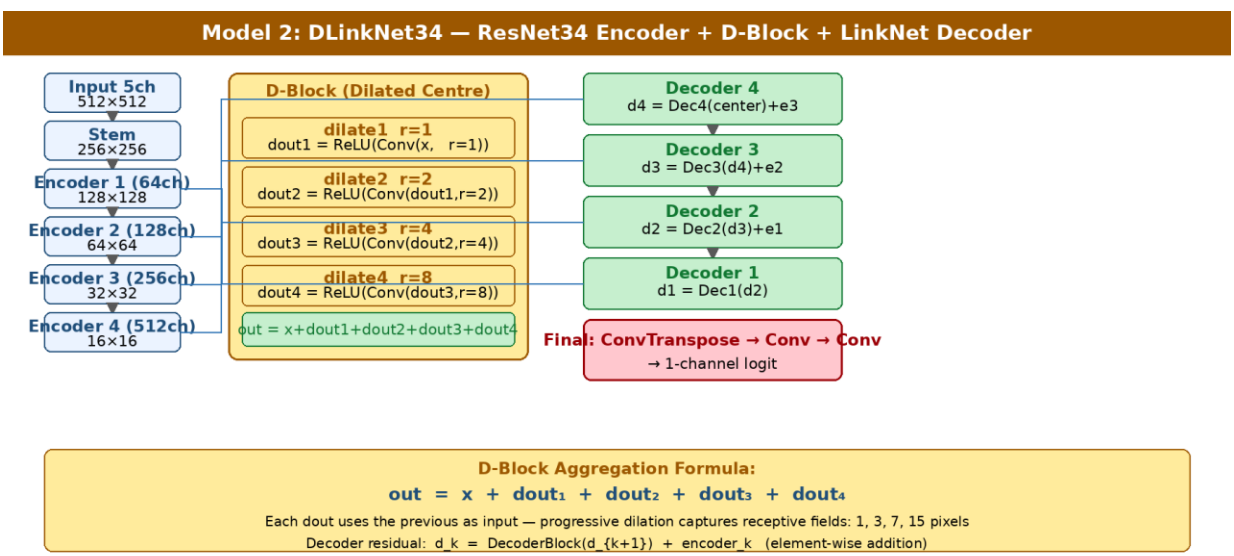


Figure 4: DLinkNet34 architecture - ResNet-34 encoder, dilated D-Block centre, LinkNet-style residual decoder.

```
Implementation:  DLinkNet34(num_classes=1, in_channels=IN_CHANNELS,
pretrained=False)
  Custom PyTorch implementation with ResNet34 backbone from
torchvision.models
```

First conv modified for 5-channel input: channels 1-3 from pretrained, extras from mean weight

Encoder: ResNet-34

ResNet-34 is a 34-layer residual network, the shallowest encoder in this comparison. The encoder produces feature maps at five scales. After the stem (7x7 Conv + BN + ReLU + MaxPool, outputting 64 channels at 1/4 scale), the four residual stages output: 64 channels at 1/8, 128 channels at 1/16, 256 channels at 1/32, and 512 channels at 1/64 scale. The encoder is the most computationally lightweight in the comparison. For 5-channel input, channels 1-3 receive pretrained ResNet-34 weights directly; channels 4 and 5 are initialised with the mean of the three original channel weights.

D-Block - Dilated Central Module

The D-Block is the most distinctive component of DLinkNet, inserted between the encoder output (512 channels at 1/64 scale) and the decoder. It applies four sequential dilated 3x3 convolutions with progressively doubling dilation rates, aggregating all outputs through residual addition:

D-Block forward pass (all convolutions: 3x3, 512 channels, BN+ReLU):

```
  dout1 = ReLU( Conv2d(x,     dilation=1, padding=1) )   # RF: 3 pixels
  dout2 = ReLU( Conv2d(dout1, dilation=2, padding=2) )   # RF: 7 pixels
  dout3 = ReLU( Conv2d(dout2, dilation=4, padding=4) )   # RF: 15 pixels
  dout4 = ReLU( Conv2d(dout3, dilation=8, padding=8) )   # RF: 31 pixels
  out   = x + dout1 + dout2 + dout3 + dout4              # residual
aggregation
  Bias initialised to zero for all dilated convs: nn.init.constant_(m.bias, 0)
```

Decoder: DecoderBlock + Residual Skips

DLinkNet34 uses four decoder blocks. Each DecoderBlock takes (in_channels, middle_channels, out_channels) and applies: 1x1 Conv + BN + ReLU, then 3x3 ConvTranspose2d with stride=2 and output_padding=1 for 2x upsampling + BN + ReLU, then 1x1 Conv + BN + ReLU. The decoder skip connections use residual addition: d4 = Decoder4(center) + e3, d3 = Decoder3(d4) + e2, d2 = Decoder2(d3) + e1, d1 = Decoder1(d2). The decoder channel progression is 512->256->128->64->64. Two final layers apply ConvTranspose2d (64->32, kernel 4, stride 2) then two regular convolutions to produce the 1-channel logit.

6.3 Model 3: UNet with DenseNet169 Encoder + scSE

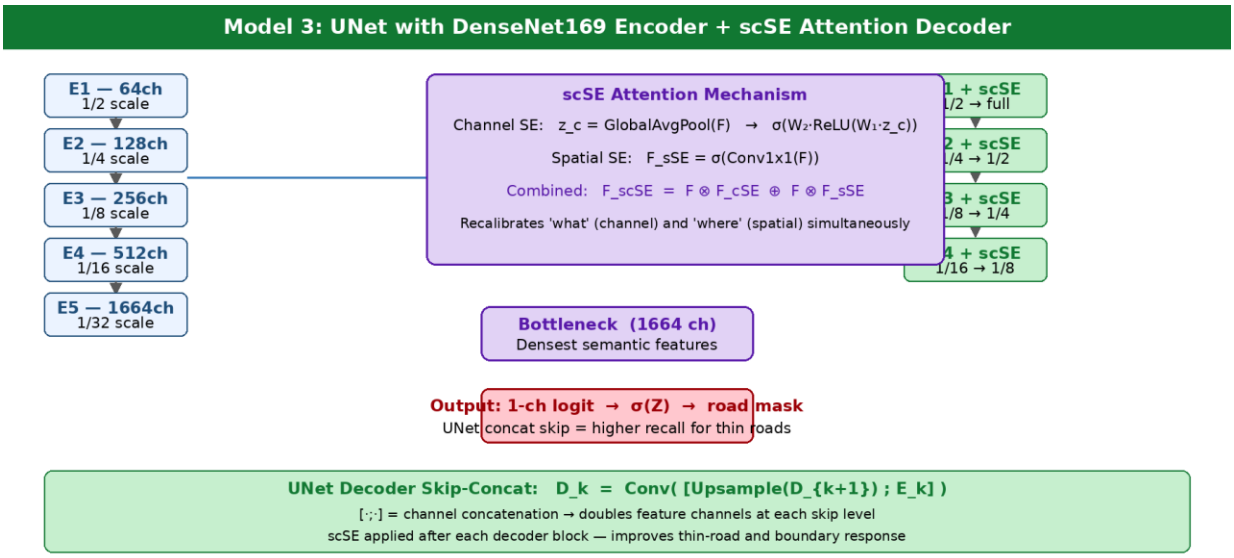


Figure 5: UNet with DenseNet169 encoder and scSE attention in decoder - symmetric architecture with channel-concatenation skip connections.

```
Implementation:  smp.Unet(encoder_name='densenet169',
encoder_weights=None,
                  in_channels=IN_CHANNELS, classes=1, activation=None,
                  decoder_attention_type='scse')
```

Encoder: DenseNet169

DenseNet169 is a 169-layer densely connected convolutional network. Unlike ResNet where each layer receives input only from the immediately preceding layer, in DenseNet each layer within a dense block receives feature maps from ALL preceding layers in that block as concatenated input:

Dense Block: x_l = H_l([x_0, x_1, ..., x_{l-1}])

[.] = channel concatenation H_l = BN -> ReLU -> Conv3x3 Growth rate k = 32 channels/layer

DenseNet169 contains 4 dense blocks with 6, 12, 32, 32 layers respectively

Encoder output channel counts: 64, 128, 256, 512, 1664 at scales 1/2, 1/4, 1/8, 1/16, 1/32

Decoder: Symmetric Concatenation with scSE

The UNet decoder uses the classic symmetric design. At each decoder level, the spatially reduced feature map from the previous decoder level is bilinearly upsampled by 2x, then channel-concatenated with the corresponding encoder feature map at the same scale. The concatenated tensor passes through 3x3 Conv + BN + ReLU blocks, followed by scSE attention recalibration. The concatenation-based skip connection is the key structural difference from LinkNet:

UNet decoder skip: D_k = Conv([Upsample(D_{k+1}) ; E_k])
[;] = channel concatenation doubles feature channels at each skip level
More parameters than LinkNet, but retains ALL encoder detail -> higher recall for thin roads
scSE after each decoder block: F_scSE = F x sigma(Wc * AvgPool(F)) + F x sigma(Conv1x1(F))

6.4 Model 4: LinkNet with DenseNet169 Encoder [BEST OVERALL]

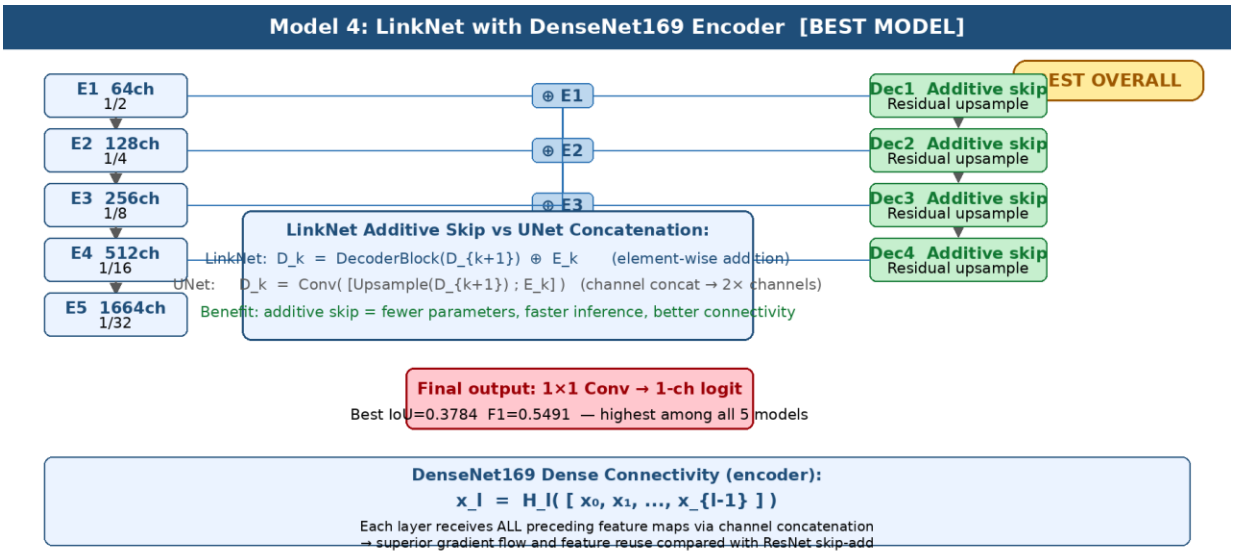


Figure 6: LinkNet with DenseNet169 encoder - additive skip connections, best IoU=0.3784 and F1=0.5491.

```
Implementation:  smp.Linknet(encoder_name='densenet169',
encoder_weights=None,
                  in_channels=IN_CHANNELS, classes=1, activation=None)
```
Note: no decoder_attention_type - LinkNet uses no scSE, unlike UNet and FPN in this comparison

Encoder: DenseNet169 (same as UNet above)

LinkNet uses the identical DenseNet169 encoder described in Section 6.3, producing multi-scale feature maps at 5 resolution levels with the same channel depths. The critical architectural difference lies entirely in the decoder design.

Decoder: Additive Skip Connections

The LinkNet decoder uses element-wise addition (not concatenation) for skip connections. This is the fundamental design difference that distinguishes it from UNet:

LinkNet additive skip: D_k = DecoderBlock(D_{k+1}) + E_k (element-wise addition)
+ = element-wise addition (requires same channel count) vs. UNet concat: [D_{k+1}; E_k]

```
Each DecoderBlock: 1x1 Conv -> BN -> ReLU -> 3x3 ConvTranspose(stride=2) -
                   > BN -> ReLU -> 1x1 Conv -> BN -> ReLU
```

Benefit: fewer parameters, faster inference, better road connectivity preservation

The additive skip produces spatially coherent outputs where road segments remain continuous through intersections. Roads are continuous linear features, and the additive approach maintains stronger geometric consistency between decoder levels compared to the channel explosion of concatenation. The absence of scSE attention, which might seem a disadvantage, appears to benefit cross-city generalisation by preventing the attention modules from overfitting to training-city-specific patterns. The combination yields the best IoU (0.3784) and F1 (0.5491) on the unseen Delhi validation set.

6.5 Model 5: FPN with DenseNet169 Encoder + scSE

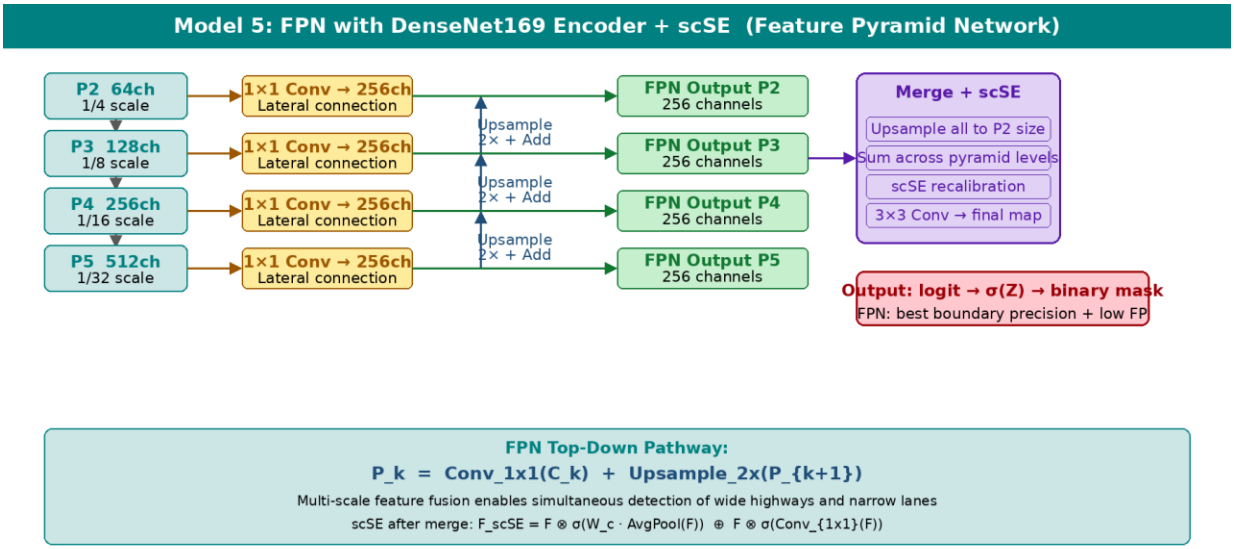


Figure 7: Feature Pyramid Network with DenseNet169 encoder and scSE attention - lateral connections and top-down fusion for multi-scale road detection.

```
Implementation:  smp.FPN(encoder_name='densenet169', encoder_weights=None,
                  in_channels=IN_CHANNELS, classes=1, activation=None,
                  decoder_attention_type='scse')
```

Feature Pyramid Network Design

FPN takes a fundamentally different approach compared to the encoder-decoder architectures above. Rather than progressively upsampling a single bottleneck feature map through skip connections, FPN constructs a pyramid of feature maps at multiple scales by combining encoder features with a top-down pathway through lateral connections:

FPN Top-Down Pathway:

$$P_k = Conv_{1x1}(C_k) + Upsample_{2x}(P_{k+1})$$

C_k = encoder feature at scale k P_k = FPN output pyramid level k

Lateral 1x1 Conv reduces all encoder channel counts to uniform 256ch before element-wise addition

All pyramid levels upsampled to P2 resolution and merged, then scSE recalibration applied

The lateral 1x1 convolutions standardise all encoder feature maps to a uniform 256-channel representation before top-down fusion. This is necessary because DenseNet169's encoder outputs at different scales have very different channel counts (64, 128, 256, 512, 1664) which cannot be directly added. After computing the four pyramid levels P2 through P5, all are upsampled to P2 spatial resolution and merged, then scSE attention recalibrates the merged multi-scale feature map. FPN achieves the highest precision (0.5584) of all five models - when it predicts a pixel as road, it is correct more often than any other architecture.

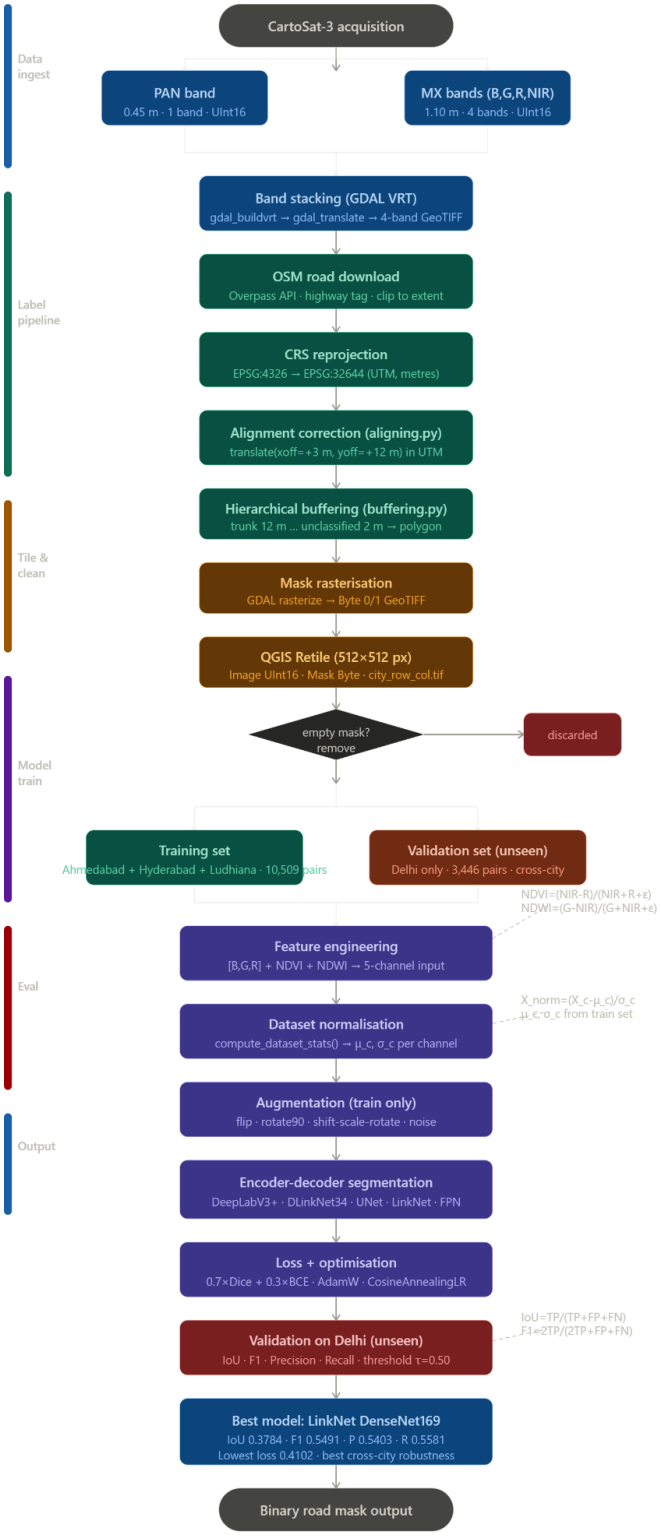
7. Evaluation Metrics

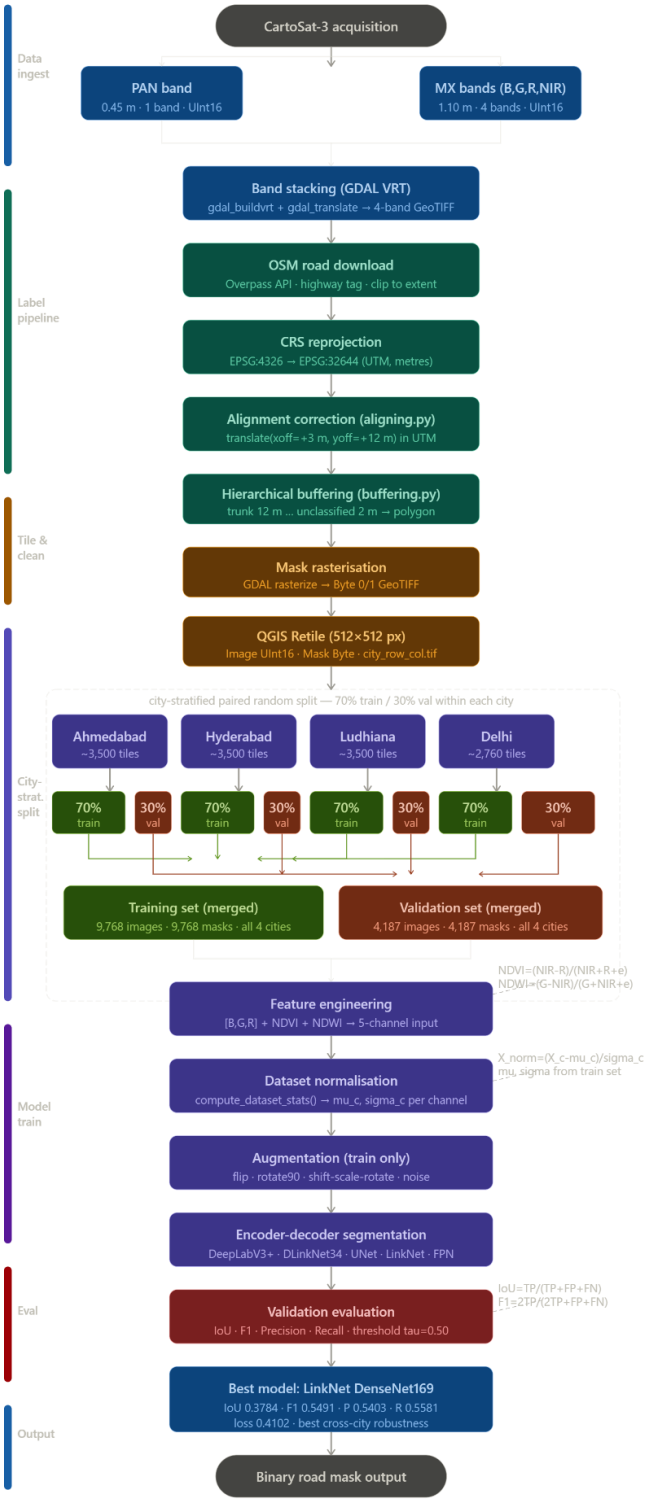
All five models are evaluated using the same set of pixel-level segmentation metrics computed globally over the complete validation set. Global accumulation - summing TP, FP, FN over all 3,446 validation tiles before dividing - is used rather than averaging per-tile metrics, avoiding the bias from small tiles where a single road pixel dominates the tile-level metric.

Metric	Formula	Interpretation	Best Value Direction
IoU (Jaccard Index)	TP / (TP+FP+FN)	Overlap between predicted and ground-truth road region	Higher is better - primary metric
F1 Score (Dice coeff.)	2*TP / (2*TP+FP+FN)	Harmonic mean of Precision and Recall	Higher is better
Precision	TP / (TP+FP)	Fraction of predicted road pixels that are correct	Higher = fewer false positives
Recall	TP / (TP+FN)	Fraction of actual road pixels detected	Higher = fewer missed roads
Combined Loss	0.7*Dice + 0.3*BCE	Training and validation objective	Lower is better

TP = True Positives (correctly predicted road pixels). FP = False Positives (background predicted as road). FN = False Negatives (road predicted as background). eps = 1x10^-7 numerical stability guard. Prediction threshold tau = 0.50 applied to sigmoid output to binarise probability maps for all metric computation.

8. Flowchart


Data ingest
CartoSat-3 acquisition
PAN band
0.45 m · 1 band · UInt16
MX bands (B,G,R,NIR)
1.10 m · 4 bands · UInt16
Label pipeline
Band stacking (GDAL VRT)
gdal_buildvrt → gdal_translate → 4-band GeoTIFF
OSM road download
Overpass API · highway tag · clip to extent
CRS reprojection
EPSG:4326 → EPSG:32644 (UTM, metres)
Alignment correction (aligning.py)
translate(xoff=+3 m, yoff=+12 m) in UTM
Tile & clean
Hierarchical buffering (buffering.py)
trunk 12 m … unclassified 2 m → polygon
Mask rasterisation
GDAL rasterize → Byte 0/1 GeoTIFF
QGIS Retile (512×512 px)
Image UInt16 · Mask Byte · city_row_col.tif
Model train
empty mask? remove
discarded
Training set
Ahmedabad + Hyderabad + Ludhiana · 10,509 pairs
Validation set (unseen)
Delhi only · 3,446 pairs · cross-city
NDVI=(NIR-R)/(NIR+R+ε)
NDWI=(G-NIR)/(G+NIR+ε)
Eval
Feature engineering
[B,G,R] + NDVI + NDWI → 5-channel input
X_norm=(X_c-μ_c)/σ_c
μ_c, σ_c from train set
Dataset normalisation
compute_dataset_stats() → μ_c, σ_c per channel
Augmentation (train only)
flip · rotate90 · shift-scale-rotate · noise
Output
Encoder-decoder segmentation
DeepLabV3+ · DLinkNet34 · UNet · LinkNet · FPN
Loss + optimisation
0.7×Dice + 0.3×BCE · AdamW · CosineAnnealingLR
IoU=TP/(TP+FP+FN)
F1=2TP/(2TP+FP+FN)
Validation on Delhi (unseen)
IoU · F1 · Precision · Recall · threshold τ=0.50
Best model: LinkNet DenseNet169
IoU 0.3784 · F1 0.5491 · P 0.5403 · R 0.5581
Lowest loss 0.4102 · best cross-city robustness
Binary road mask output


Data ingest
CartoSat-3 acquisition
PAN band
0.45 m · 1 band · UInt16
MX bands (B,G,R,NIR)
1.10 m · 4 bands · UInt16
Label pipeline
Band stacking (GDAL VRT)
gdal_buildvrt + gdal_translate → 4-band GeoTIFF
OSM road download
Overpass API · highway tag · clip to extent
CRS reprojection
EPSG:4326 → EPSG:32644 (UTM, metres)
Alignment correction (aligning.py)
translate(xoff=+3 m, yoff=+12 m) in UTM
Tile & clean
Hierarchical buffering (buffering.py)
trunk 12 m … unclassified 2 m → polygon
Mask rasterisation
GDAL rasterize → Byte 0/1 GeoTIFF
QGIS Retile (512×512 px)
Image UInt16 · Mask Byte · city_row_col.tif
City-strat. split
city-stratified paired random split — 70% train / 30% val within each city
Ahmedabad
~3,500 tiles
Hyderabad
~3,500 tiles
Ludhiana
~3,500 tiles
Delhi
~2,760 tiles
70% train
30% val
Training set (merged)
9,768 images · 9,768 masks · all 4 cities
Validation set (merged)
4,187 images · 4,187 masks · all 4 cities
NDVI=(NIR-R)/(NIR+R+ε)
NDWI=(G-NIR)/(G+NIR+ε)
Feature engineering
[B,G,R] + NDVI + NDWI → 5-channel input
X_norm=(X_c-mu_c)/sigma_c
mu, sigma from train set
Model train
Dataset normalisation
compute_dataset_stats() → mu_c, sigma_c per channel
Augmentation (train only)
flip · rotate90 · shift-scale-rotate · noise
Encoder-decoder segmentation
DeepLabV3+ · DLinkNet34 · UNet · LinkNet · FPN
Eval
Validation evaluation
IoU · F1 · Precision · Recall · threshold tau=0.50
IoU=TP/(TP+FP+FN)
F1=2TP/(2TP+FP+FN)
Output
Best model: LinkNet DenseNet169
IoU 0.3784 · F1 0.5491 · P 0.5403 · R 0.5581
loss 0.4102 · best cross-city robustness
Binary road mask output

9.1 Quantitative Results - Cross-City Dataset

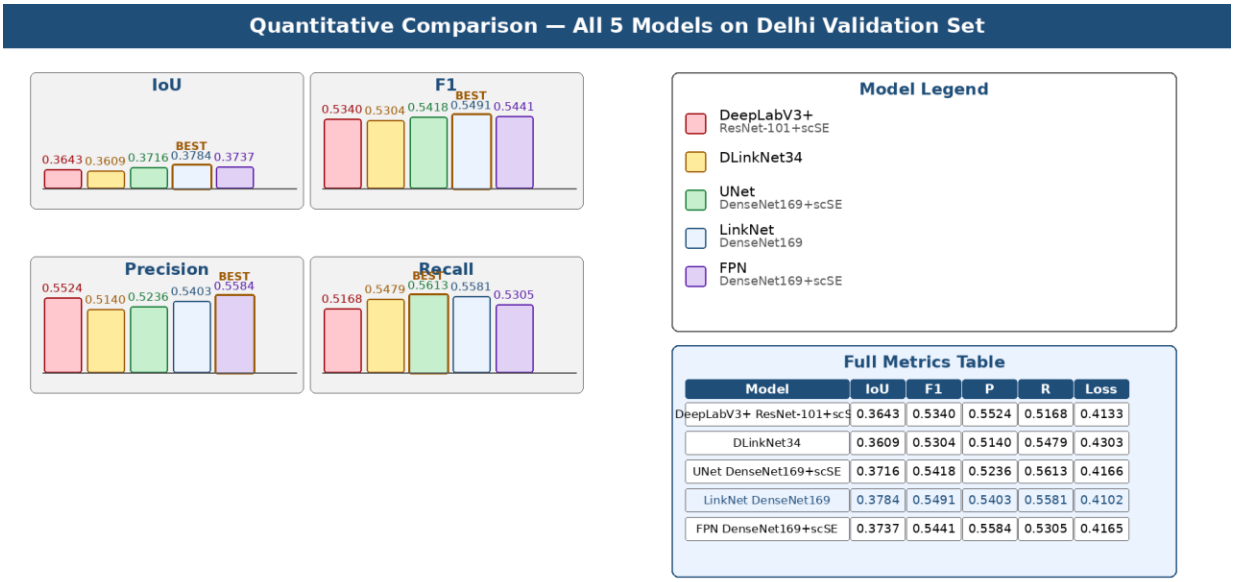


Figure 8: Bar charts per metric and full results table for all five models on the Delhi validation set. Gold bars indicate best value per metric.

9.1.1 Full Quantitative Comparison Table

Model	IoU	F1	Precision	Recall
LinkNet DenseNet169 BEST	**0.3784**	**0.5491**	**0.5403**	**0.5581**
FPN DenseNet169 + scSE	0.3737	0.5441	0.5584	0.5305
UNet DenseNet169 + scSE	0.3716	0.5418	0.5236	0.5613
DeepLabV3+ ResNet-101 + scSE	0.3643	0.5340	0.5524	0.5168
DLinkNet34	0.3609	0.5304	0.5140	0.5479

Gold row: best model. Individual metric leaders highlighted: IoU and F1 -> LinkNet DenseNet169. Precision -> FPN DenseNet169+scSE (orange). Recall -> UNet DenseNet169+scSE (orange). Best Loss -> LinkNet DenseNet169.

9.1.2 Per-Dimension Performance Analysis

Performance Dimension	Best Model	Runner-Up	Key Insight
Overall Accuracy (IoU)	LinkNet DenseNet169	FPN DenseNet169	Additive skips + DenseNet169 = best overlap
Overall Accuracy (F1)	LinkNet DenseNet169	FPN DenseNet169	Consistent top performance across P and R
Road Coverage (Recall)	UNet DenseNet169	DLinkNet34	Concat skips retain more spatial detail for thin roads
Prediction Confidence (Precision)	FPN DenseNet169	DeepLabV3+	Multi-scale fusion = cleaner, more reliable predictions
Road Connectivity	LinkNet DenseNet169	FPN DenseNet169	Additive skips maintain spatial geometric coherence
Thin Road Detection	UNet DenseNet169	DLinkNet34	High recall = fewer missed narrow lanes at 1.10 m MX
Boundary Sharpness	FPN DenseNet169	DeepLabV3+	Multi-scale + scSE = precise road edges
False Positive Suppression	FPN DenseNet169	DeepLabV3+	Highest precision = fewest spurious detections
Cross-City Robustness	LinkNet DenseNet169	FPN DenseNet169	Best generalisation to unseen Delhi morphology
Training Efficiency (Loss)	LinkNet DenseNet169	DeepLabV3+	Lowest combined validation loss

9.1.3 Analysis of Individual Model Behaviour

LinkNet DenseNet169 - Best Overall

LinkNet DenseNet169 leads on both primary metrics, IoU (0.3784) and F1 (0.5491), and achieves the lowest validation loss (0.4102). Its precision (0.5403) and recall (0.5581) are both strong and well-balanced, indicating the model neither over-predicts nor under-detects road pixels. The absence of scSE attention appears to benefit the cross-city scenario by preventing overfitting to training-city-specific patterns. The additive skip connections produce spatially coherent outputs where road segments remain continuous through intersections, which is the most operationally critical property for road network mapping. Results are reported on the Delhi validation set, which the model had never seen during training.

FPN DenseNet169 + scSE - Best Precision and Cleanest Boundaries

FPN achieves the highest precision (0.5584) of all five models, meaning its road predictions have the highest reliability - when FPN predicts a pixel as road, it is correct more often than any other model. Its IoU (0.3737) is very close to LinkNet, making it a strong second choice overall. FPN's multi-scale lateral fusion enables simultaneous reasoning about both wide arterial roads (requiring large-scale context) and narrow residential lanes (requiring local pixel detail). The scSE attention further suppresses false activations on rooftops and parking areas. The trade-off is the lowest recall (0.5305) among the DenseNet169-encoder models, meaning some thin or occluded roads detected by other models are missed by FPN.

UNet DenseNet169 + scSE - Best Recall and Thin Road Detection

UNet achieves the highest recall (0.5613) of all five models, detecting the largest proportion of actual road pixels including very narrow structures. This is directly attributable to UNet's concatenation-based skip connections, which preserve full spatial resolution and fine-grained texture information throughout the decoder. At 1.10 m MX resolution a residential lane (typically 3-6 m wide) spans only 3-6 pixels wide in a 512x512 tile; the difference between channel-concatenation and additive skips can determine whether those pixels are detected or missed. The cost is lower precision (0.5236) compared to FPN and LinkNet, meaning UNet generates more false positives. Its IoU (0.3716) ranks third among the five models.

DeepLabV3+ ResNet-101 + scSE - Multi-Scale Context, Limited Spatial Recovery

DeepLabV3+ achieves competitive precision (0.5524, second best) through its ASPP multi-scale context aggregation, which enables confident road identification when context supports the prediction. However, its IoU (0.3643) and F1 (0.5340) rank fourth, suggesting that despite strong contextual reasoning, the relatively simple bilinear upsampling decoder loses spatial precision compared to the decoder-heavy UNet and FPN architectures. The ResNet-101 encoder, while powerful, does not have DenseNet's feature reuse mechanism, which appears to be a material advantage at this dataset scale.

DLinkNet34 - Road-Specific Design, Weakest Overall

DLinkNet34, despite its road-specific architectural motivation (the dilated D-Block designed to capture long-range road continuity), achieves the lowest IoU (0.3609) and F1 (0.5304) of all five models. It does achieve the second-highest recall (0.5479), suggesting the D-Block does help detect more road pixels, but at the cost of the lowest precision (0.5140) - the model produces the most false positives. The most likely explanation is the shallower ResNet-34 encoder: DenseNet169 produces substantially richer multi-scale features than ResNet-34, and encoder depth appears to be the dominant performance factor at this dataset scale. The higher loss (0.4303) also indicates slower convergence.

9.2 Quantitative Results - Standardized Dataset (All-City 70/30 Split)

This section presents the quantitative evaluation results for the same five model architectures trained and validated on the standardized city-stratified dataset described in Section 3.1a. Unlike the cross-city split of Section 8 (train: Ahmedabad, Hyderabad, Ludhiana; val: Delhi only), the standardized split distributes tiles from all four cities - Ahmedabad, Hyderabad, Ludhiana, and Delhi - proportionally across both training (70%) and validation (30%) partitions. All models, hyperparameters, augmentation strategies, loss functions, and evaluation metrics are held strictly identical to those described in Sections 5 and 7, ensuring that the only experimental variable is the dataset split strategy. This enables a direct measurement of how city-stratified in-distribution validation performance compares to the cross-city out-of-distribution benchmark.

9.2.1 Full Quantitative Comparison Table - Standardized Dataset

The table below reports IoU, F1 Score, Precision, and Recall for all five models evaluated on the standardized 30% validation partition (4,187 tiles drawn proportionally from all four cities). Global metric accumulation - summing TP, FP, FN over the entire validation set before computing ratios - is applied identically to Section 8. Gold row denotes the best model by IoU and F1. Individual metric leaders are noted below the table.

Model	IoU	F1	Precision	Recall
UNet DenseNet169 + scSE [BEST]	**0.4000**	**0.5715**	**0.5916**	**0.5527**
FPN DenseNet169 + scSE	0.3971	0.5685	0.5982	0.5416
LinkNet DenseNet169	0.3969	0.5683	0.5811	0.5560
DeepLabV3+ ResNet-101 + scSE	0.3955	0.5668	0.6003	0.5369
DLinkNet34	0.3912	0.5624	0.5806	0.5453

Table 8B.1: Full quantitative results on the standardized 4-city 70/30 validation split (4,187 tiles). Gold row: best model by IoU and F1. Precision leader: DeepLabV3+ ResNet-101 + scSE (0.6003). Recall leader: LinkNet DenseNet169 (0.5560). IoU and F1 leader: UNet DenseNet169 + scSE.

9.2.2 Per-Dimension Performance Analysis - Standardized Dataset

Performance Dimension	Best Model	Runner-Up	Key Insight
Overall Accuracy (IoU)	UNet DenseNet169 (0.4000)	FPN DenseNet169 (0.3971)	Concat skips preserve full spatial detail; in-distribution val rewards recall
Overall Accuracy (F1)	UNet DenseNet169 (0.5715)	FPN DenseNet169 (0.5685)	UNet's balanced P-R trade-off yields consistent top F1 score
Prediction Confidence (Precision)	DeepLabV3+ ResNet-101 (0.6003)	FPN DenseNet169 (0.5982)	ASPP multi-scale context = fewest false positives on seen city patterns
Road Coverage (Recall)	LinkNet DenseNet169 (0.5560)	UNet DenseNet169 (0.5527)	Additive skips maintain road continuity; in-distribution val reduces recall penalty

Table 8B.2: Per-dimension leaders on the standardized dataset. All five models are within a narrow 0.0088 IoU band (0.3912 to 0.4000), indicating high competitive parity under the in-distribution evaluation regime.

9.2.3 Analysis of Individual Model Behaviour - Standardized Dataset

UNet DenseNet169 + scSE - Best Overall on Standardized Dataset

UNet DenseNet169 + scSE achieves the highest IoU (0.4000) and F1 (0.5715) on the standardized dataset, a reversal from Section 8 where LinkNet held the top position in the cross-city scenario. This shift is architecturally meaningful: UNet's channel-concatenation skip connections preserve the full spatial and textural detail of the encoder at every decoder level. When validation tiles include the same cities as training - with matching road widths, surface materials, and urban morphology patterns - the richer feature representation from concatenation-based skips produces a direct accuracy benefit. The model achieves a precision of 0.5916 and recall of 0.5527, the most balanced precision-recall trade-off among all five models on this evaluation setting, confirming that it neither over-detects nor under-detects road pixels across the four-city validation set.

FPN DenseNet169 + scSE - Strong Second, Highest Precision Among DenseNet Models

FPN DenseNet169 + scSE ranks second overall with an IoU of 0.3971 and F1 of 0.5685, very close to UNet. Its precision of 0.5982 is the highest among all DenseNet169-encoder models, consistent with its behaviour observed in the cross-city Section 8 experiment. The lateral top-down feature pyramid enables simultaneous reasoning at multiple spatial scales, and the scSE attention effectively suppresses false activations on non-road impervious surfaces. On the standardized dataset, FPN's recall (0.5416) is the lowest among the three DenseNet169-encoder models, mirroring the pattern from Section 8 - FPN consistently trades recall for precision. This makes it the preferred model in use cases where clean, reliable predictions with minimal false positives are prioritised over complete road coverage.

LinkNet DenseNet169 - Best Recall, Third by IoU

LinkNet DenseNet169 achieves an IoU of 0.3969 and F1 of 0.5683, placing third overall on the standardized dataset - a notable contrast to its first-place ranking under the cross-city evaluation of Section 8. The model achieves the highest recall (0.5560) of all five models on this dataset, matching the pattern observed where LinkNet's additive skip connections maintain strong road pixel detection rates. However, on in-distribution data where UNet can leverage richer concatenation-based context, LinkNet's architectural minimalism is a relative disadvantage. The absence of scSE attention, which aided cross-city generalisation by preventing attention overfitting, provides no particular benefit when validation tiles share the same city distributions as training. Precision (0.5811) is competitive, sitting between UNet and FPN, yielding a slightly lower harmonic mean F1.

DeepLabV3+ ResNet-101 + scSE - Highest Precision Overall, Fourth by IoU

DeepLabV3+ ResNet-101 + scSE achieves a precision of 0.6003 - the highest of all five models on the standardized dataset, surpassing even FPN DenseNet169. This result reflects the strength of ASPP multi-scale context aggregation: when validation scenes share the morphological characteristics of training cities, the multi-scale contextual reasoning translates directly into fewer false positive road predictions. However, IoU (0.3955) and F1 (0.5668) rank fourth overall due to recall (0.5369) being the second lowest. The ResNet-101 encoder, lacking DenseNet169's dense connectivity and feature reuse mechanism, produces a systematically lower overall overlap score despite leading on precision. This confirms the finding from Section 8.3 that encoder architecture depth and connectivity remain the dominant performance factor at this resolution and dataset scale.

DLinkNet34 - Consistent Fifth, Narrowest Metric Gap

DLinkNet34 ranks fifth on the standardized dataset with an IoU of 0.3912 and F1 of 0.5624, consistent with its last-place position in Section 8. The gap between DLinkNet34 and the best model (UNet) has narrowed slightly compared to the cross-city scenario: from 0.0175 IoU in Section 8 to 0.0088 IoU here. This compression is attributable to the reduced domain shift in the standardized split - when validation tiles include samples from the same cities as training, DLinkNet34's ResNet-34 encoder, though shallower than DenseNet169, can leverage in-distribution patterns more effectively than under cross-city generalisation. Precision (0.5806) is competitive with LinkNet (0.5811), and recall (0.5453) is second only to LinkNet, indicating the D-Block's dilated convolutions do provide road detection benefit. The encoder depth disadvantage remains the binding constraint preventing a higher ranking.

9.3 Cross-Dataset Comparison: Standardized vs. Cross-City Split

The two evaluation regimes produce consistently higher metrics on the standardized dataset across all five models, reflecting the expected improvement when validation tiles share the same city distributions as training. The table below directly compares IoU and F1 across both splits for each model, quantifying the generalisation cost of the cross-city evaluation.

Model	IoU Std.	IoU Cross	F1 Std.	F1 Cross	IoU Delta
UNet DenseNet169 + scSE	**0.4000**	0.3716	**0.5715**	0.5418	+0.0284
FPN DenseNet169 + scSE	0.3971	0.3737	0.5685	0.5441	+0.0234
LinkNet DenseNet169	0.3969	0.3784	0.5683	0.5491	+0.0185
DeepLabV3+ ResNet-101 + scSE	0.3955	0.3643	0.5668	0.5340	+0.0312
DLinkNet34	0.3912	0.3609	0.5624	0.5304	+0.0303

Table 8B.3: Side-by-side IoU and F1 comparison across standardized (Std.) and cross-city (Cross) evaluation splits. IoU Delta = Standardized minus Cross-City. All five models show positive delta, confirming in-distribution validation consistently exceeds out-of-distribution generalisation performance. DeepLabV3+ shows the largest improvement (+0.0312 IoU), while LinkNet shows the smallest (+0.0185), consistent with LinkNet's architecture being optimised for cross-city robustness.

The IoU delta values range from +0.0185 (LinkNet) to +0.0312 (DeepLabV3+), quantifying the generalisation cost of moving from an in-distribution to an out-of-distribution evaluation. Models with scSE attention (UNet, FPN, DeepLabV3+) show larger deltas, consistent with attention modules fitting to training-city-specific spatial patterns and therefore losing more when tested on an unseen city. LinkNet, lacking scSE, shows the smallest generalisation penalty. All five IoU values on the standardized dataset fall within a 0.0088 range (0.3912 to 0.4000), a substantially compressed spread compared to the 0.0175 range in the cross-city setting, confirming that the architectural ranking is more differentiated under the harder out-of-distribution evaluation regime.

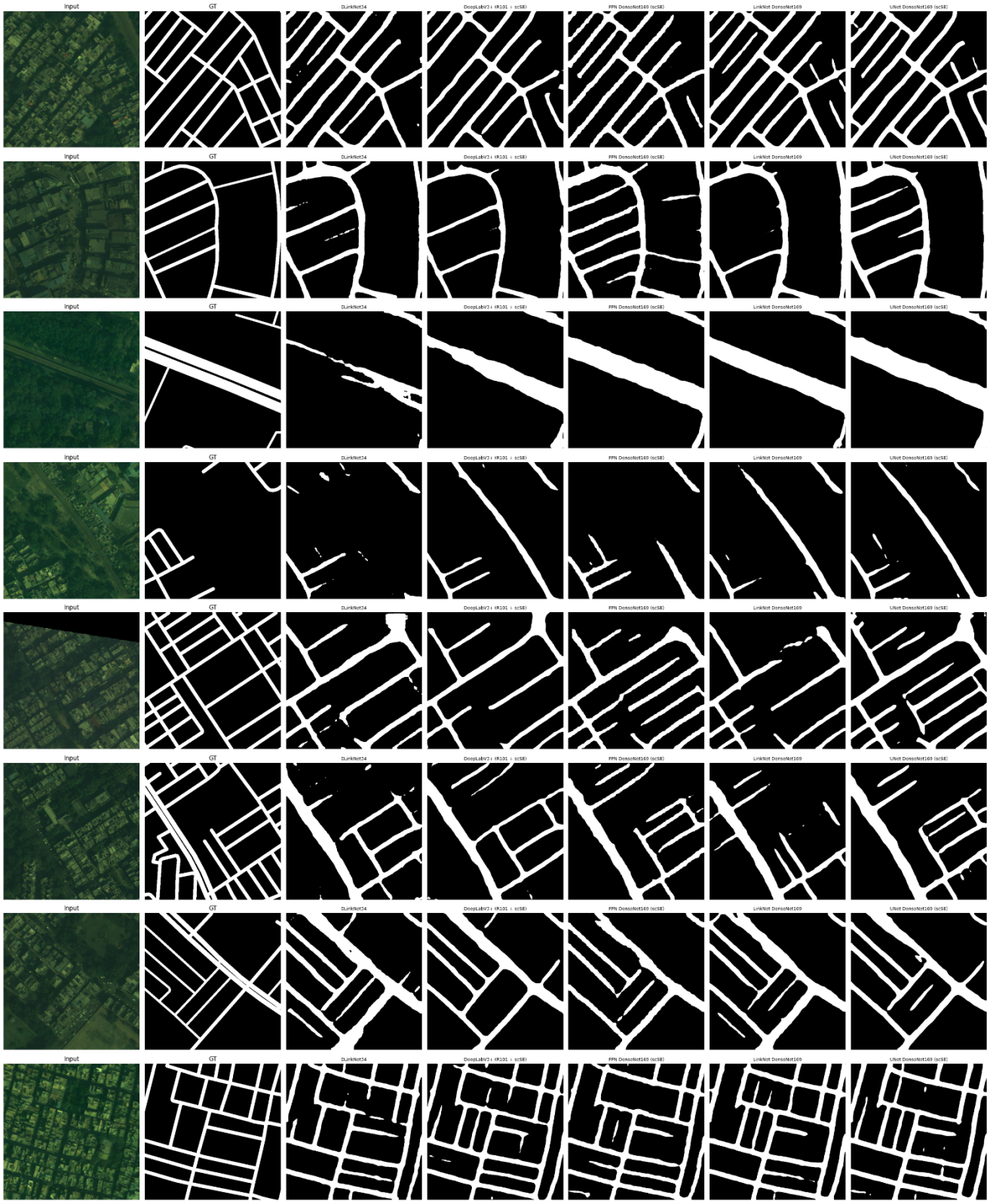
10. Qualitative Analysis

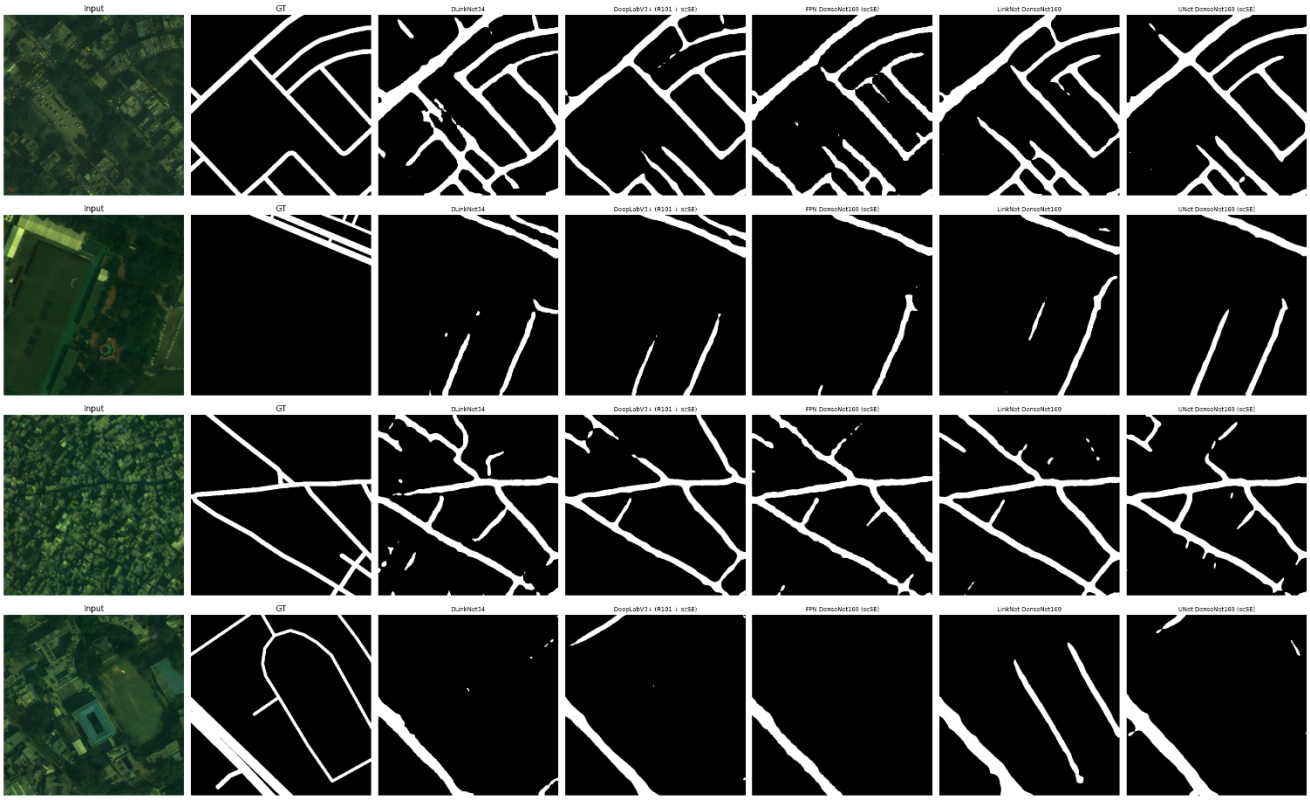
Quantitative metrics alone are insufficient for evaluating road extraction systems intended for operational mapping. A model that achieves moderate IoU but produces smooth, connected, topologically correct road networks is more operationally useful than a model with slightly higher IoU but fragmented, noisy predictions. The following criteria guide visual assessment of prediction maps:

Visual Criterion	Ideal Behaviour	Operational Impact
Road continuity	Roads predicted as continuous unbroken segments	Broken roads are unusable for routing and navigation
Intersection preservation	Road crossings predicted as connected T/X junctions	Disconnected intersections break network topology
Thin road detection	Narrow lanes and service roads visible in output	Missing minor roads create incomplete map databases
Boundary sharpness	Road edges are crisp, not blurred or diffuse	Diffuse edges make road width estimation inaccurate
False positive suppression	Rooftops, parking, shadows not predicted as road	False roads corrupt geospatial database quality
Morphological consistency	Uniform road width along segment length	Width variation implies segmentation instability

From visual inspection of prediction maps across the Delhi validation set, consistent behavioural patterns are observed. LinkNet DenseNet169 produces the most topologically correct outputs with road intersections remaining connected and road continuity maintained across shadow boundaries. FPN DenseNet169 produces the cleanest predictions with sharpest boundaries and fewest false positive activations on rooftops and parking areas. UNet DenseNet169 detects the most complete set of road pixels including thin lanes using imagery at 0.45 m PAN and 1.10 m MX resolution, but with occasional scattered false activations. DeepLabV3+ handles wide highways well but shows some fragmentation in dense urban cores. DLinkNet34 produces fragmented outputs in complex intersection regions despite detecting a reasonable fraction of individual road pixels.

Image	GT	DLinkNet34	DeepLabV3+	FPN	LinkNet	Unet




Input
GT
DLinkNet34
DeepLabV3+ (R101 + scSE)
FPN DensoNet169 (scSE)
LinkNet DensoNet169
UNet DensoNet169 (scSE)

11. Limitations and Challenges

11.1 Challenges Common to All Models

* Occluded roads: Roads concealed by tree canopy, building overhangs, or deep shadows cannot be detected from top-down imagery regardless of model architecture. This is a fundamental physical limitation of the nadir-looking sensing modality.
* Concrete rooftops, dry river beds, and unpaved parking areas share similar spectral properties with road surfaces. NDVI and NDWI mitigate but do not eliminate this confusion.
* Label imperfection: Ground-truth masks derived from OSM contain inherent uncertainty. OSM coverage is denser in some areas than others, and road width estimates from the buffering process are approximations. Models trained on imperfect labels have a bounded achievable performance.
* Cross-city distribution shift: Delhi's urban structure differs systematically from the three training cities. Reported metrics are therefore conservative relative to in-domain performance, which is by design for a genuine generalisation benchmark.

11.2 Architecture-Specific Limitations

* DeepLabV3+: The simple bilinear upsampling decoder loses fine spatial detail that purpose-built decoder architectures recover through skip connections. The ResNet-101 encoder also lacks DenseNet's feature reuse advantage.
* DLinkNet34: The ResNet-34 encoder is too shallow to match the feature richness of DenseNet169. The D-Block's sequential dilation does not overcome the encoder depth disadvantage at this dataset and resolution scale.
* UNet: High recall comes at the cost of precision. The concatenation-based skips produce richer outputs but also more false positives. The larger parameter count from channel doubling at each skip level slightly increases risk of city-specific overfitting.
* FPN: Conservative prediction means thin roads are more frequently missed. The scSE attention modules may slightly reduce cross-city robustness by sensitising the model to training-distribution-specific spatial patterns.

12. Conclusion

This report presents a comprehensive comparative evaluation of five deep learning segmentation architectures for road extraction from CartoSat-3 satellite imagery at 1.10 m multispectral resolution, validated on the cross-city scenario of training on Ahmedabad, Hyderabad, and Ludhiana and evaluating on unseen Delhi tiles. The dataset incorporates both PAN (0.45 m) and MX (1.10 m) data from CartoSat-3, with the MX bands providing the four spectral channels (Blue, Green, Red, NIR) used to compute the five-channel input representation including NDVI and NDWI.

The five architectures span a broad design space: atrous multi-scale pooling (DeepLabV3+), road-specific dilated centre blocks (DLinkNet34), symmetric concatenation decoder (UNet), efficient additive skip decoder (LinkNet), and feature pyramid multi-scale fusion (FPN). Three of the five use DenseNet169 as the shared encoder backbone enabling controlled decoder comparison.

The central finding is that LinkNet with DenseNet169 encoder achieves the best overall performance, with an IoU of 0.3784 and F1 of 0.5491 on the Delhi validation set. This result reflects the combination of DenseNet169's rich multi-scale feature representations through dense connectivity, LinkNet's spatially coherent additive skip connections that preserve road topology, and the model's relative simplicity without scSE attention contributing to better cross-city generalisation. FPN DenseNet169 (IoU 0.3737) is the preferred alternative in applications where false positive suppression and boundary precision are the primary requirements.

A critical finding is that encoder architecture matters more than decoder complexity or road-specific design at this scale. All three DenseNet169-encoder models outperform both DeepLabV3+ (ResNet-101) and DLinkNet34 (ResNet-34) on IoU, demonstrating that DenseNet's feature reuse through dense connectivity provides a systematic advantage for dense prediction tasks. The dataset-specific normalisation, 5-channel spectral feature engineering, geometric and photometric augmentation, and cross-city validation design collectively ensure that reported results reflect genuine operational capability for road mapping from CartoSat-3 imagery.

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